

APPLICATION

1. Solve the following equations using logarithm laws. Write as a single logarithm.

$$6^{3x+1} = 2^{2x+3}$$

$$\log(6^{3x+1}) = \log(2^{2x+3})$$

$$(3x+1) \log 6 = (2x+3) \log 2$$

$$3x \log 6 + \log 6 = 2x \log 2 + 3 \log 2$$

$$3x \log 6 - 2x \log 2 = 3 \log 2 - \log 6$$

$$\rightarrow x(3 \log 6 - 2 \log 2) = 3 \log 2 - \log 6$$

$$x = \frac{3 \log 2 - \log 6}{3 \log 6 - 2 \log 2}$$

$$x = \frac{\log \frac{2^3}{6}}{\log \frac{6^3}{2^2}}$$

$$x = \log_{\frac{6^3}{2^2}} \left(\frac{2^3}{6} \right)$$

2. The mass of a radioactive substance is 500g from the beginning and only 60g remains after 40 days. Determine the half life, to one decimal place:

$$f(t) = 500 \left(\frac{1}{2} \right)^{\frac{t}{t_{1/2}}}$$

sub in pt (60, 40)

$$60 = 500 \left(\frac{1}{2} \right)^{\frac{40}{t_{1/2}}}$$

$$0.12 = \left(\frac{1}{2} \right)^{\frac{40}{t_{1/2}}}$$

THINKING

1. If $\left(\frac{1}{2} \right)^{x+y} = 16$ and $\log_{x-y} 8 = -3$, calculate the values of x and y . Leave answers in exact form.

$$\log \left(\frac{1}{2} \right)^{x+y} = \log 16$$

$$(x+y) = \frac{\log 16}{\log \frac{1}{2}} = -4$$

$$\frac{\log 8}{\log (x-y)} = -3$$

$$\frac{1}{\log (x-y)} = -3$$

$$\frac{1}{\log (x-y)} = -3$$

$$\log (x-y) = -\frac{1}{3}$$

$$x-y = 8^{\frac{1}{3}}$$

$$x+y = -4 \quad \textcircled{1}$$

$$\textcircled{1} + \textcircled{2}$$

$$2x = -\frac{7}{2}$$

$$x = -\frac{7}{4}$$

$$\textcircled{1} - \textcircled{2}$$

$$2y = -\frac{9}{2}$$

$$y = -\frac{9}{4}$$

KNOWLEDGE

Must show all your work!!!

M/C	Total	Total
10/10	22/22	32/32

1. Express, $\sqrt[3]{81^3} \times \sqrt{27}$ with base 27.

$$\begin{aligned}
 &= 81^{\frac{3}{3}} \times 27^{\frac{1}{2}} \\
 &= (3^4)^{\frac{3}{3}} \times 27^{\frac{1}{2}} \\
 &= 3^{\frac{12}{3}} \times 27^{\frac{1}{2}}
 \end{aligned}$$

$$\begin{aligned}
 &= (27^{\frac{4}{3}})^{\frac{3}{3}} \times 27^{\frac{1}{2}} \\
 &= 27^{\frac{4}{3} \times \frac{3}{3}} \times 27^{\frac{1}{2}} \\
 &= 27^{\frac{12}{3}} \times 27^{\frac{1}{2}}
 \end{aligned}$$

2. Solve for x, round to 1 d.p if necessary.

a) $\log(x^2 - x - 6) - \log(3x + 6) = \log\left(\frac{5}{3}\right)$

$$\log(x^2 - x - 6) = \log\left(\frac{5}{3}\right) + \log(3x + 6)$$

$$\log(x^2 - x - 6) = \log\left(-\frac{5}{3}(3x + 6)\right)$$

$$x^2 - x - 6 = -5x - 10$$

$$x^2 + 4x + 4 = 0$$

$$(x + 2)^2 = 0$$

$$x = -2$$

$$2^x - 4 + 3(2^x) = 0$$

$$2 - 4(2^x) + 3 = 0$$

$$(2^x - 3)(2^x - 1) = 0$$

$$2^x = 3$$

$$\log 2 = \log 3$$

$$x = \log_2 3$$

$$x = 1.6$$

$$2^x = 1$$

$$\log 2 = \log 1$$

$$x = \frac{\log 2}{\log 2}$$

$$x = 0$$

b) $\log_{32} \sqrt{x^2 - 3x} = \frac{1}{5}$

$$\sqrt{x^2 - 3x} = 32^{\frac{1}{5}}$$

$$x^2 - 3x = 2$$

$$x^2 - 3x - 4 = 0$$

$$(x - 4)(x + 1) = 0$$

$$x = 4 \quad x = -1$$

$$x > -1$$

$$b) \log_3(x + 4) + \log_3(2x - 3) = \log_3 6$$

$$\log_3[(x + 4)(2x - 3)] = \log_3 6$$

$$(x + 4)(2x - 3) = 6$$

$$2x^2 - 3x + 8x - 12 = 6$$

$$2x^2 + 5x - 18 = 0$$

$$(2x + 9)(x - 2) = 0$$

$$x = -\frac{9}{2} \quad \text{or} \quad x = 2$$

$$\text{Since } x > \frac{3}{2}$$

$$x = 2$$